

Body Pointing, Acquisition and Tracking for Small Satellite Laser Communication

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ABSTRACT

Free-space optical communications in space offer many benefits over established radio frequency based communication links; in particular, high beam directivity results in efficient power usage. Such a reduced power requirement is particularly appealing to small satellites with strict size, weight and power (SWaP) requirements. In the case of free-space optical communication, precise pointing, acquisition and tracking (PAT) of the incoming beam is necessary to close the communication link. Due to the narrow beam of the laser, the critical task of accomplishing PAT becomes increasingly arduous and often requires complex systems of optical and processing hardware to account for relative movement of the terminals. Recent developments in body pointing mechanisms have allowed small satellites to point with greater precision. In this work, we consider an approach to a low-complexity PAT system that utilizes a single quad-cell photodetector as an optical spatial sensor, and exploits the body pointing capabilities of the spacecraft to perform the tracking maneuvers, eschewing the need for additional dedicated optical hardware. We look at the PAT performance of this approach from a systems analysis viewpoint and present preliminary experimental results. In particular, we examine the implementation of the system on NASA's TeraByte InfraRed Delivery (TBIRD) demonstration.

Keywords: free-space optical communication, low-Earth orbit, cubesat, body pointing, pointing acquisition and tracking

1. INTRODUCTION

Optical communication in space is attractive due to several benefits over established radio frequency based communication links. By using lasers, a much wider band of modulation frequencies is available, opening up the possibility for high rate data communications using the largely unregulated frequencies in the terahertz spectrum. In addition, laser beams have high directivity and narrow beams, and thus are particularly desirable due to efficient power delivery. In small satellites, where low size, weight and power (SWaP) is critical, optical communication offers clear advantages.

However, the cost of the narrow laser beam is an increased pointing and tracking performance requirement. Optical communication requires line of sight in order to close the communication link; pointing and tracking ensures that adequate power is received at the terminal during the communication period. Power loss at the receiving terminal results in poor signal to noise ratio and reduces the achievable communication rate. Typically, a system must be able to point to within a fraction of the beamwidth to close the link. For optical communication systems, beamwidths of ~ 10 to $1000\mu\text{rad}$ are typical.

Such space-based optical communication links have been successfully demonstrated before, such as in the Lunar Laser Communication Demonstration (LLCD) in 2013, with a downlink data rate of up to 622 Mbps¹ from the Moon and the Aerospace Corporation's Optical Communication and Sensor Demonstration (OCSD) in 2017, with a downlink data rate of 100 Mbps from a 450km low-Earth orbit.² However, LLCD, and most space-based optical communications systems since then were unable to fully take advantage of the high data rates possible with laser-based communication due to long link distances and other design constraints. At the same time, the long link distances imposed strict pointing requirements, and drove the implementation of designs

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using complex steering optics to ensure the pointing requirements could be met. Generally, components such as gimbals and other fast steering actuators have been used to achieve fine pointing performance. However, additional actuators further strain the power and weight constraints of missions.

Currently, science missions in space face a bottleneck in data transfer from LEO orbit to ground, where scientists can process and interpret data gathered. Current sensors are capable of generating terabytes of data in a short period of time, but communication links have not kept up with the advancement, rendering the use of high resolution and high rate sensing systems more challenging to realize, since the data cannot be easily recovered. The development of a low complexity and accessible solution would be useful for the science community and could enable novel research in space.

NASA's TeraByte InfraRed Delivery (TBIRD) system is one such solution currently in development. TBIRD seeks to demonstrate high downlink data rate transfer at 200 Gbps from a CubeSat in low-Earth orbit (LEO) to a ground terminal while minimizing complexity, in part by utilizing commercial off-the-shelf (COTS) parts.³ Furthermore, TBIRD uses only the spacecraft bus reaction-wheel-based actuation for pointing. The geometry of orbits in LEO result in short passes over a ground terminal several times per day, during which line of sight may be achieved. Though the passes may be on the order of a few minutes in duration, over the course of 24 hours the short bursts of communication can result in delivery of many terabytes per day from a small space terminal in LEO to a small ground terminal.³ This direct-to-Earth concept is well suited for science missions, where latency in data transmission is acceptable.

In this work, we present a simultaneous spatial tracking and communication system using a quadrant photodetector sensor, to be used as the uplink receiver on the TBIRD payload.³ The quadrant sensor uses the optical uplink to provide the spacecraft bus with pointing feedback for fine attitude control, unlike previously realized systems where pointing feedback and communication were completely decoupled. The dual-purpose uplink receiver reduces SWaP while improving pointing performance. The remainder of this work will focus on the pointing feedback aspect of the uplink receiver.

This paper will be organized as follows. In Section 2, we provide relevant background regarding TBIRD, and present the requirements of the coupled uplink system. A modulation scheme for the uplink communication is given. In Section 3, we present the uplink architecture and motivate the relevant hardware components. In Section 4, we model the behavior of the system and show how pointing feedback from the payload improves pointing error. We also show the effect of the improved pointing on the downlink pointing loss. Finally, in Section 5, we describe our experimental setup and present preliminary results.

2. SYSTEM OVERVIEW

The TBIRD uplink system is motivated by a desire to minimize complexity, size, weight and power by combining the requirements of two distinct subsystems - uplink communications and pointing, acquisition and tracking - into a single uplink receiver. This section will describe the needs of each subsystem and motivate the implementation of the uplink receiver.

TBIRD includes an uplink communication between the ground terminal and the spacecraft. In order to achieve a high data rate downlink at 200 Gbps, TBIRD implements an automatic repeat request (ARQ) protocol to ensure the reliable transmission of data in the presence of atmospheric fading. The ARQ commands are transmitted from the ground terminal to the spacecraft on the uplink when the ground terminal detects corruption or loss of a downlink data frame.⁴ Although the downlink data rate is necessarily very high, the uplink data rate carrying ARQ commands is not bound to the same requirements and can in fact be much lower; TBIRD utilizes a target uplink rate of 5kbps, which is sufficient to avoid throughput penalties associated with the ARQ protocol.⁵ The uplink together with the downlink create a communication feedback loop, as shown in the upper portion of Figure 1, between the space terminal payload and the ground terminal. The receiver architecture handling the uplink data must have sufficient bandwidth to accommodate the frequency content and modulation of the communication signal. In addition, the optical receiver sensor must be sensitive enough to detect signals at weak power levels, with a favorable signal to noise ratio since the uplink signal will experience atmospheric fading effects.

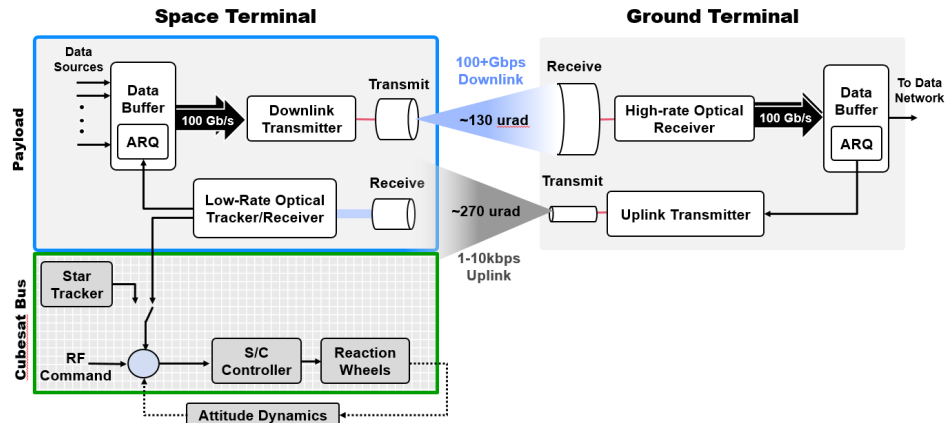


Figure 1: Block diagram of TBIRD system. The space terminal is divided into payload (blue) and CubeSat bus (green). Attitude feedback inputs into spacecraft controller can come from either an onboard star tracker or the payload optical tracker/receiver.

Pointing, acquisition and tracking (PAT) refers to the process by which the two terminals in an optical link make contact, detect the respective beams and begin to track. Because TBIRD seeks to reduce complexity by eliminating excess actuation elements such as fast steering mirrors on the payload, it relies on the CubeSat bus's reaction wheel-based attitude control for fine pointing and tracking.

Three-axis attitude control of small satellites has improved significantly due to miniaturization of star trackers and reaction wheels. Recently, the Jet Propulsion Laboratory's ASTERIA mission demonstrated $2.5 \mu\text{rad}$ RMS pointing accuracy with a two-stage control system.⁶ The Aerospace Corporation's Optical Communication and Sensor Demonstration (OCS D) mission showed that a lasercom link can be established on a small satellite using open-loop body pointing. The beam widths used on OCS D were 0.05° ($873 \mu\text{rad}$) FWHM and the spacecraft achieved about $\pm 0.01^\circ$ ($175 \mu\text{rad}$) pointing accuracy.² In comparison, to achieve high burst rates on TBIRD, a narrow downlink beam of about $130 \mu\text{rad}$ is necessary, which places an even tighter pointing requirement on the spacecraft.

On TBIRD, a quad sensor and ground beacon are utilized to improve body pointing with closed-loop feedback. For acquisition, the spacecraft must first open-loop point using information from the onboard star tracker accurately enough that the ground beacon is within the field of view of the quad sensor, approximately 20 mrad . Once the beacon is acquired, closed-loop feedback is provided by the payload to the bus at a rate of 30 Hz to improve pointing to a fraction of the downlink beamwidth.

However, TBIRD couples the pointing and communication subsystems together through a common uplink receiver. Spatial tracking feedback is generated from a communication signal, rather than from a CW or modulated beacon. A single quad sensor on the spacecraft is used to receive the uplink signal. The sum of the quadrants' received power yields the communication signal, and comparison of the quadrants' received power yields the pointing feedback that is later provided to the bus for closed-loop pointing.

In some optical communication systems, such as LLCD, an uplink beacon is modulated in order to aid in background rejection.^{7,8} Without modulation, stray light can impede beacon detection. For example, the Optical Payload for Lasercomm Science (OPALS) demonstration in 2014 reported challenges in determining appropriate beacon detection thresholds due to background distribution uncertainties.⁹ Beacon modulation at a known frequency helps mitigate concerns about background light, since filtering for the modulated frequency can improve detection. However, the frequency spectrum of a communication signal may not have any discrete tones to track on. Therefore, careful choice of modulation for the uplink signal is required.

In order to accommodate the tracking's need for a known frequency content, a binary pulse position modulation (PPM) scheme with pulse shaping is used. Each symbol is divided into two slots, wherein the location of the slot in which the pulse is located determines the value of the symbol. Dead time is inserted on either side of the pulse within the slot, which increases the peak power of the received signal, and introduces a discrete tone

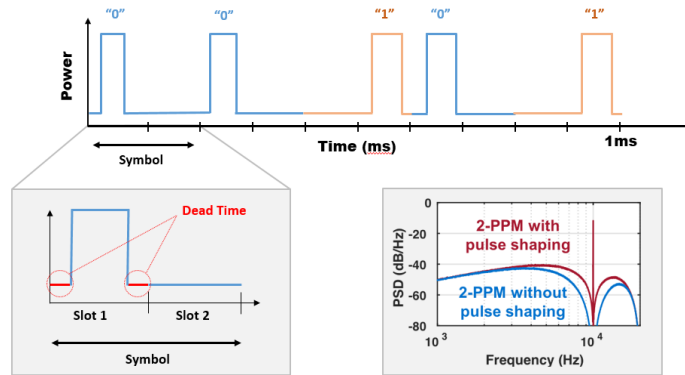


Figure 2: Uplink modulation scheme. Insertion of dead time in a binary PPM waveform with a 10 kHz slot rate introduces discrete tone at 10kHz for tracking.

which can later be isolated through a narrow bandpass filter, as shown in Figure 2. Because the communication aspect of the signal requires much greater bandwidth (5kbps data rate) while the pointing aspect of the signal requires much lower bandwidth (30Hz updates) owing to mechanical actuation constraints and the response time of the attitude control, it is possible to track on the pulse-shaped signal.

In considering the system, we include possible challenges such as atmospheric fading and power fluctuations, which can cause deterioration of the uplink signal in situations where the signal to noise ratio is poor. When received power decreases, the noise can overpower the uplink receiver and hinder signal processing attempts to recover pointing feedback and communication signals. For the remainder of this paper, we will focus on the hardware, simulation and testing of the pointing feedback aspect of the dual-purpose uplink receiver.

3. HARDWARE DESIGN

The hardware for the TBIRD uplink receiver was chosen to minimize SWaP and complexity, while balancing the requirements of both the uplink communication signal and pointing feedback.

Fundamentally, the priorities of the two subsystems are different; while comm prioritizes high sampling bandwidth to enable higher possible data rates, the attitude determination goal of the pointing feedback aspect prioritizes minimization of background and increased field of view. Furthermore, we must choose COTS hardware that will not only serve both subsystems, but is also reasonable to implement. This section will focus on the chosen sensor and computation components for TBIRD.

In this application a quadrant photodetector sensor was chosen as the exclusive sensor for the uplink receiver. A block diagram detailing the uplink receiver components and architecture is shown in Figure 3. The quadrant sensor has four photodetectors arranged in a 2x2 grid. Each quadrant produces an analog output commensurate with the amount of power received on the quadrant. An amplification circuit amplifies the signals and an analog to digital converter (ADC) digitizes the signals to be processed. The sensor, along with the amplification and digitization circuits are pictured in Figure 4. The small footprint of the sensor and circuit is well suited for TBIRD's small payload volume. Figure 4b additionally shows the relative positioning of the receive aperture and the quad sensor itself.

The quadrant detector was chosen over a larger focal plane array primarily due to the low implementation complexity and straightforward interpretation of the four signals. A focal plane array would suit the uplink tracking by improving background rejection, but it has a much higher interfacing and computational complexity due to the very high frame rate that is required to support 5 kbps comm.

Because the uplink receiver must interface with both the communications system (internal to the payload) and the tracking system (external to the payload) while only using a single sensor, signal processing must occur to obtain the data for each subsystem; the communication data must be demodulated, and the meaningful pointing

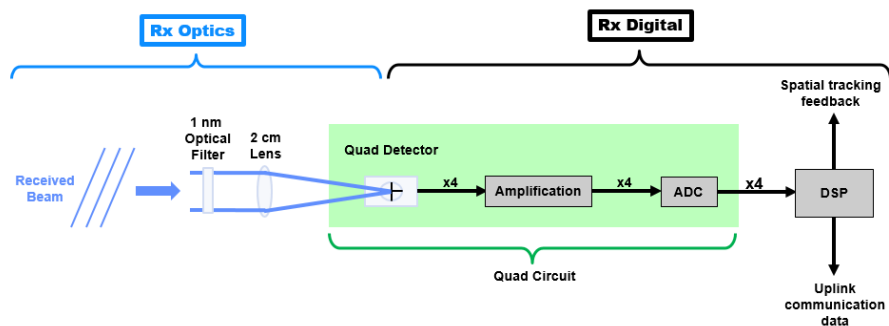
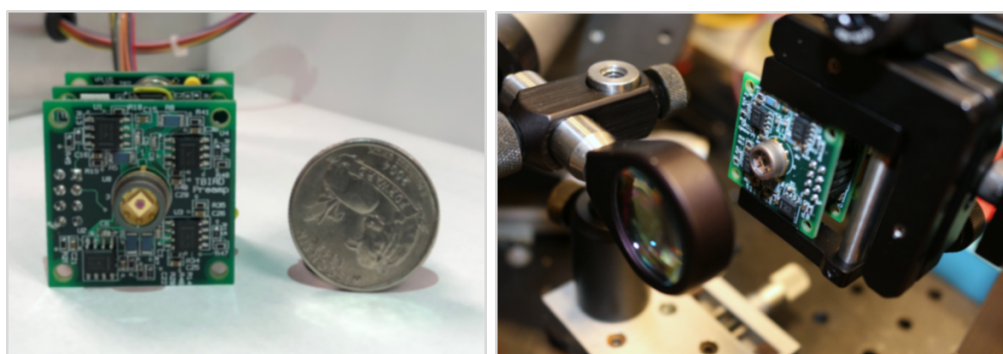


Figure 3: Block diagram of optical and digital components. The receive optics are shown on the left side of the diagram in blue, while the digital components are shown on the right side in black. The green highlighted area comprises the quad sensor circuit. The circuit amplifies and digitizes the quad sensor signals for processing.



(a) Size relative to U.S. quarter (b) Mounted in front of receive aperture.
Figure 4: Quadrant sensor with amplification and digital conversion circuit.

feedback measurements must be determined. To minimize complexity, the signal processing is performed in the digital domain.

A COTS radiation hardened microcontroller was chosen to perform the tracking part of the signal processing for the digitized output of the quadrant sensor. Although microcontrollers are not as computationally efficient as a field-programmable gate array (FPGA), and the chosen microcontroller is not specifically optimized for digital signal processing (DSP) applications, they require much less power. However, microcontrollers are not capable of executing parallel computations, require several clock cycles to perform computations or read/write to memory, which makes the efficient design and implementation of DSP a challenge. Here, the choice of the quadrant sensor proves sensible, because the computational resources required to interface with sensor is still within the capabilities of the chosen microcontroller.

In the TBIRD architecture, it is desirable to reduce the amount of time that the high power components are powered up, due to thermal considerations. Excess heat generated by components such as an existing onboard FPGA limits the duration of the downlink transmissions. Thus, the design of the TBIRD space terminal utilizes the microcontroller to perform those signal processing operations associated with acquisition and tracking, which occurs prior to downlink transmissions, and the onboard FPGA to demodulate the uplink data. As a result, it is possible to perform tracking without powering on the FPGA.

4. SIMULATION

A time-domain simulation was developed in Simulink to characterize the spacecraft pointing performance. A block diagram of the major components is shown in Figure 5. A 6-U CubeSat is modeled in a 400 km orbit. A reference trajectory is generated for a ground station pass with a maximum elevation of 75°. The spacecraft is

assumed to have one star tracker and gyroscopes in three axes, as well as three reaction wheels. These sensors and actuators are modeled based on existing COTS components. Attitude estimation is performed with a 6-state extended Kalman filter (EKF) and a simple proportional-derivative controller generates reaction wheel commands. The control gains are selected to set the closed-loop control bandwidth at 0.5 Hz with no payload feedback and 1 Hz with payload feedback. Sources of error in the simulation include uncertainty in inertial properties, reaction wheel misalignment, sensor noise, sensor misalignment, gyro drift, and onboard timing error.

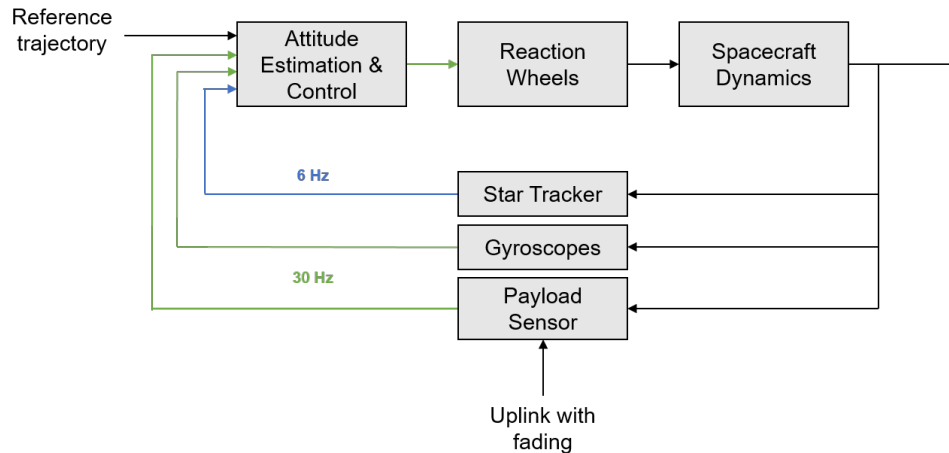


Figure 5: Block diagram of spacecraft pointing simulation.

Feedback from the communications payload is provided to the bus at a rate of 30 Hz. A binary PPM waveform with a 10 kHz slot rate and with 50% dead time per slot is generated to simulate the uplink. The dead time introduces a 10 kHz tone that is used for background rejection. An atmospheric fading profile is applied to the waveform. On the spacecraft, a phase-locked loop is used to demodulate the uplink in software and generate an angular attitude measurement. The measured angular error is fed directly into the controller and the EKF provides an error estimate for the remaining axis.

Results from a simulated spacecraft pass are shown in Figure 6. If no feedback from the payload is provided, the root-mean-square pointing error of the bus is 62 μrad . With payload feedback, the pointing error reduces to 22 μrad RMS. The payload feedback greatly reduces low-frequency bias since the quad cell is rigidly coupled with the downlink transmitter.

This result highlights a key advantage to the single uplink receiver design. Because the pointing error is determined directly from the uplink receiver which is in turn rigidly coupled with the downlink transmitter, the pointing acquisition and tracking system is robust to potential misalignment between the payload and the star tracker on the spacecraft bus. Although misalignment is not included in the simulation at this time, it is reasonable to predict that with misalignment, the pointing performance with the payload sensor would not significantly differ from the aligned case, while the pointing performance without the payload sensor (relying only on the spacecraft bus pointing capabilities) could be potentially seriously degraded.

Figure 7 relates these pointing errors back to the communications link losses. The losses in Fig. 7 correspond with the pointing errors with payload feedback in Fig. 6. There is a large initial transient as the filters converge. The median pointing loss is 0.2 dB and losses are less than 1.3 dB 99% of the pass. These pointing losses are low enough to support a 200 Gbps burst rate.

5. EXPERIMENTAL RESULTS

5.1 Experimental Setup

An experimental setup was created for the testing and verification of the TBIRD uplink receiver. A key goal of the experimental setup was to create a setting which would reasonably reflect on-orbit conditions. The experimental setup, pictured in Figure 8 includes the quadrant sensor circuit and microcontroller for testing. A

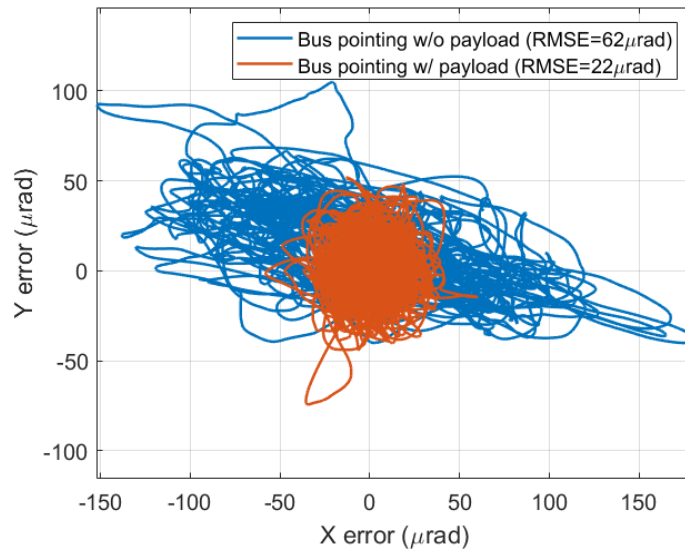


Figure 6: Simulated spacecraft pointing error for a 160 second pass from 30° elevation through a maximum of 75° elevation. Spacecraft bus pointing without payload feedback (blue) and with payload feedback (orange) are shown.

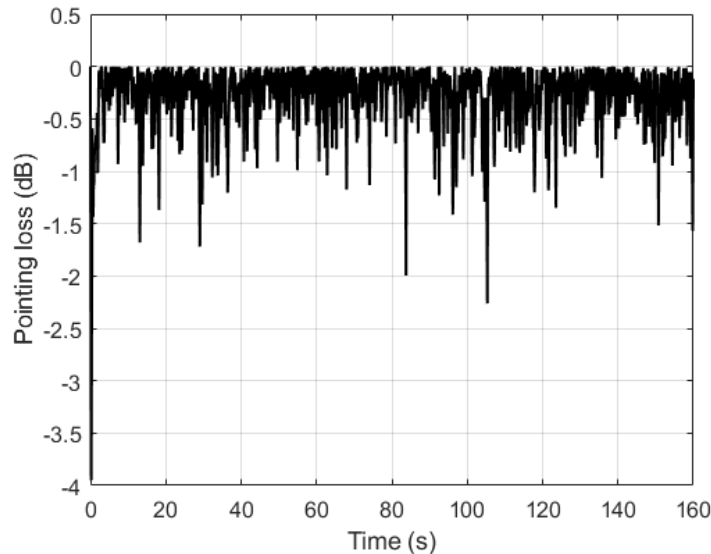


Figure 7: Downlink pointing loss with payload feedback. The median loss is 0.2 dB, and the 99th percentile is 1.3 dB. There is an initial transient period of increased loss.

fiber collimator for the laser was sized such that the beam blanketed a 2cm aperture lens, which will serve as TBIRD's space terminal receive aperture. The setup was verified with a beam profiler, which checked that the beam was appropriately sized and reasonably suited our expectations. In addition, a fast steering mirror (FSM) was included in order to introduce tracking disturbances on the system. An autocollimator was inserted into the system to measure the tracking disturbances induced by the FSM. All parts were separately tested and qualified before testing.

The testing inputs, which consisted of the uplink signal waveform as well as the tracking disturbances were controlled with a Speedgoat Real-Time target. This allowed for a centralized testing interface. The received signal at the quadrant sensor was amplified and digitized on the sensor circuit, and the resulting digitized values

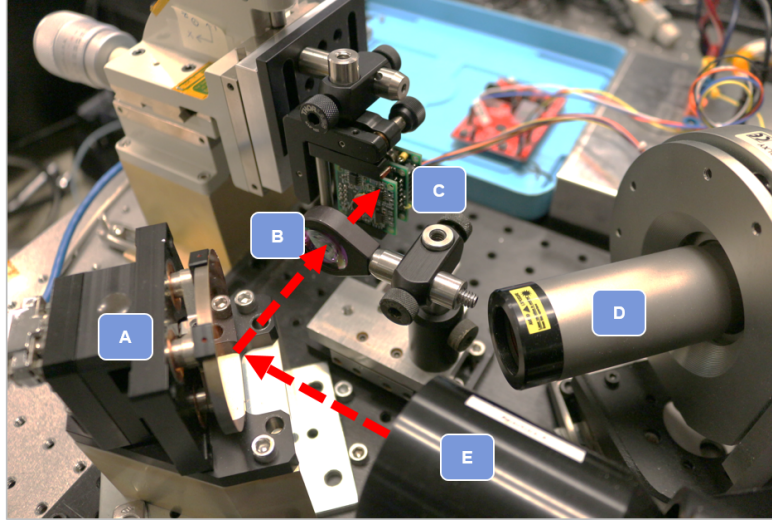


Figure 8: Experimental setup. Uplink beam path indicated by arrows.

Components legend: A: Fast-steering mirror, B: Receive aperture lens, C: Quadrant sensor circuit, D: Autocollimator, E: Fiber collimator

sampled by the microcontroller. In the work done for this paper, values sampled at 1.85 kHz were then sent to a host computer for offline signal processing. Though this received data rate was slower than the ultimate goal, it was constrained by the need for offline signal processing. In the future, the signal processing required for tracking will be fully implemented on the microcontroller, and will attain faster sampling rates.

5.2 Experimental Data

A preliminary test was conducted in which the FSM was driven in a Lissajous pattern in a high signal to noise ratio scenario. The total received power on the quadrant sensor was -58dBm. The signal from the quad circuit was sampled by the microcontroller, and the result from the offline signal processing is plotted in Figure 9.

Although the system has not undergone full calibration, the results, which show the agreement of the quad sensor measurement and the true disturbance are promising. The two-axis error is $5.7\mu\text{rad}$ RMS. After calibration, it is expected that the RMS error will be reduced.

6. CONCLUSION

This paper introduced the dual-purpose uplink receiver architecture for NASA's TeraByte InfraRed Delivery demonstration. The uplink receiver combines both the uplink communication subsystem with the pointing, acquisition and tracking subsystem. By using a single sensor and utilizing a microcontroller for the tracking signal processing, size, weight and power are conserved. A simulation including payload and spacecraft dynamics predicted the performance of the system. The result of the simulation showed that for a simulated pass, payload pointing feedback was able to reduce the pointing error from $62\mu\text{rad}$ RMS to $22\mu\text{rad}$ RMS, with a median pointing loss of 0.2 dB which is adequate to support a 200 Gbps burst rate. An experimental setup was designed and preliminary results were presented, showing the uncalibrated performance of the quad sensor tracking disturbance measurement. Without calibration, pointing feedback was able to measure disturbances introduced by a FSM with a two-axis error of $5.7\mu\text{rad}$ RMS.

Ongoing and future work will include the necessary calibration of the quad sensor, as well as the implementation of the tracking signal processing routine on the microcontroller. Thorough performance characterization will confirm the system behavior. In addition, the spacecraft dynamics simulation will be incorporated into the test bench.

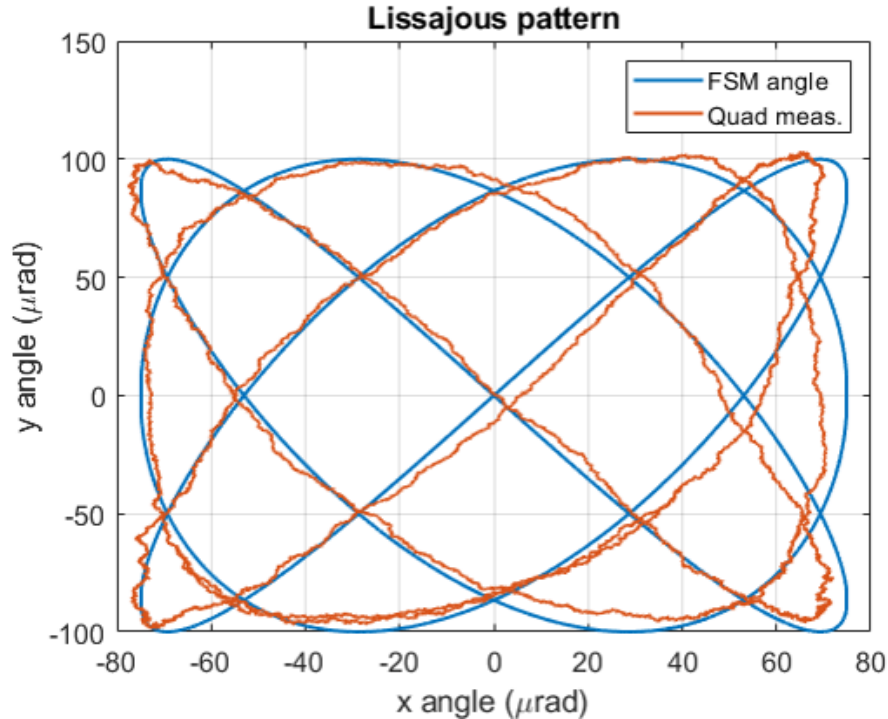


Figure 9: Experimental result. Tracking disturbances introduced with a fast steering mirror (blue) compared to quad sensor measured disturbances (orange). Two-axis error: 5.7 μrad RMS

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